

Junghwan Kim

김정환

Ph.D. Candidate, Human-Centered Computing Lab
Interdisciplinary Program in Artificial Intelligence,
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RESEARCH INTERESTS

Human-Computer Interaction; LLM-augmented interface design for domain experts; Human-AI collaboration. My work advances the thesis that *interface design, not model capability, is the primary bottleneck for expert use of large language models*, grounded in HCI theory (reification, distributed cognition, articulation work, boundary infrastructure).

EDUCATION

Seoul National University, Graduate School of Convergence Science & Technology

Ph.D. Candidate (Integrated M.S./Ph.D.), Interdisciplinary Program in Artificial Intelligence

Human-Centered Computing Lab | Advisor: Prof. Bongwon Suh | Expected graduation: Feb 2027

Dissertation (in progress): *LLM-Augmented Interface Design for Domain Experts*

Aug 2022 – Present

Seoul, Korea

Seoul National University

B.S. in Information Science & Culture; B.S. in Electrical & Computer Engineering

GPA: 3.96 / 4.3

Mar 2015 – Aug 2022

Seoul, Korea

PUBLICATIONS

(★ award-winning paper; † equal contribution / co-first author)

Peer-Reviewed — First Author

Junghwan Kim, H. Jeon, J. Lee, D. Heo, B. Suh. **CaseSense: Mapping the Legal Case Landscape with Multi-Dimensional Visual Exploration**. *Proc. 39th Annual ACM Symp. on User Interface Software and Technology (UIST)*, 2026. (accepted)

Junghwan Kim[†], K. Park[†], H. Song, B. Suh. **DEBSIM: Productive Friction for Learning Transfer in LLM-Based Argumentation Tutoring**. *Proc. ACM Conf. on Interactive Surfaces and Spaces (ITS)*, 2026. ★ **Best Full Paper Award**

Junghwan Kim[†], D. Heo[†], K. Kim, J. Lee, B. Suh. **Co-Overcooked: Cognitive Constraints on Partner Modeling in Human-AI Team Composition**. *Proc. Annual Meeting of the Cognitive Science Society (CogSci)*, 2026. (Oral presentation) ★ **Student Travel Award**

Junghwan Kim[†], H. Jeon[†], D. Heo, J. Lee, B. Suh. **LegisFlow: Enhancing Korean Legal Research with Temporal-Aware LLM Interfaces**. *Proc. 38th Annual ACM Symp. on User Interface Software and Technology (UIST)*, 2025.

Junghwan Kim[†], H. Oh[†], B. Suh. **ToneCanvas: Visually Painting Character Tone and Terms for Narrative Consistency**. *Adjunct Proc. 39th Annual ACM Symp. on User Interface Software and Technology (UIST Adjunct)*, 2026. (Poster, accepted)

Junghwan Kim[†], D. Heo[†], H. Hwang, J. Y. You, B. Suh. **Between the Clinic and the Kitchen: A Constraint RAG Chatbot as Boundary Infrastructure for Dialysis Care**. *CSCW 2026 Companion* (Poster).

Junghwan Kim, K. Park, J. Ryu, H. Song, B. Suh. **When Nurses Meet Vibe-Coding: Shifting from Problem Talk to Situated Specification in Participatory Design**. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*, 2026.

Peer-Reviewed — Co-Author

H. J. Hwang, **Junghwan Kim**, D. Heo, J. Y. You, J. Lee, S. Kwon, K. D. Yoo, J. H. Y. Park, B. Suh, J. P. Lee, K. W. Lee. **Effects of a Personalized Retrieval-Augmented Generation Chatbot on Information Needs in Chronic Kidney Disease Patients: Mixed Methods Randomized Controlled Trial**. *Interactive Journal of Medical Research*, 2026. (accepted, in press; doi:10.2196/96501)

J. Ryu, H. Song, **Junghwan Kim**, K. Park, B. Suh. **From Objects to Influences: Rethinking Deletion in Learning-Based Systems**. *CHI EA*, 2026.

K. Park, H. Song, S. Seo, **Junghwan Kim**, B. Suh. **“Ask Sir Oliver Ingham”: LLM-based Social Simulations for History Education**. *CHI EA*, 2025.

M. Shin, J. Choi, H. Song, **Junghwan Kim**, B. Suh. **“I feel you, but...”: Exploring the Boundaries of Pragmatic Markers in Real-time Voice-based Conversational Agents**. *CHI EA*, 2025.

Junghwan Kim, K. Park, S. Park, H. Kim, B. Suh. **MAS-LitEval: Multi-Agent System for Literary Translation Quality Assessment**. *arXiv:2506.14199*, 2025.

S. Seo, **Junghwan Kim**, M. Shin, B. Suh. **LLMDR: LLM-Driven Deadlock Detection and Resolution in Multi-Agent Pathfinding**. *arXiv:2503.00717*, 2025.

DOMESTIC PUBLICATIONS (KOREAN HCI SOCIETY)

김정환, 서봉원. CAT를 넘어 LAT로: LLM 번역 검수를 위한 인터페이스 Co-design. Proc. of HCI KOREA 2026, 41--49, 2026.

신민정, 최재훈, **김정환**, 송형우, 서봉원. LLM 기반 음성 어시스턴트의 한국어 담화 표지 활용을 위한 디자인 가이드라인 연구. 한국 HCI 학회 학술대회, 431--437, 2025.

송형우, 서승배, **김정환**, 김연준, 신민정, 외. 최근 5년간 장애 학생을 위한 보조 기술의 활용 탐색: 문헌고찰. 한국 HCI 학회 학술대회, 1343--1349, 2025.

이상아, **김정환**, 송형우, 서봉원. 시맨틱 문장 분석을 활용한 뉴스 인터페이스 디자인. 한국 HCI 학회 학술대회, 91--97, 2024.

HONORS & AWARDS

Yulchon AI Young Researcher Scholarship (AI+X), Nongshim Yulchon Foundation & SNU AI Institute	2026
Best Full Paper Award, ACM Interactive Surfaces and Spaces (ITS) — DEBSIM	2026
Student Travel Award, Cognitive Science Society (CogSci) — Co-Overcooked	2026
3rd Place, Big-Data Financial Investment Model (FIM) Showcase, SNU Eng. & QUANTIT	Aug 2023
Excellence Award, NICE D&B SME Closure Prediction Challenge	Dec 2022

RESEARCH & PROFESSIONAL EXPERIENCE

XAI Researcher (Contract), Qraft Technologies	Dec 2021 – Jul 2022
Explainable AI for time-series forecasting (Temporal Fusion Transformer, IMV-LSTM, DARNN, DLinear); macro-data asset allocation; launched the Eugene OCIO Fund with XAI visualization (Tableau, Streamlit).	
Lab Intern, Human-Centered Computing Lab, SNU	Jul 2021 – Dec 2021
Shopping-pattern analysis and next-purchase-date modeling with ML/DL.	
Research Intern, System Semiconductor for AI (SSAI), SNU	Mar 2021 – Jun 2021
Remote proctoring for non-face-to-face testing; on-device object detection (YOLOv5, MobileNetV2).	

TEACHING

Teaching Assistant, Samsung Electronics DS2 Program (Data Scientist Training)	2022 – 2025
Teaching Assistant, Digital Forensics — Dept. of Mathematical & Information Sciences, SNU	2024 – 2026
Mentor (2024) & Coordinator (2025), LG AI Youth Camp	2024 – 2025
Teaching Assistant, Social Computing, SNU	2024 – 2026

SKILLS & CERTIFICATIONS

GenAI & LLMs: architecting and deploying agentic, multi-agent LLM systems and RAG pipelines at scale; OpenAI, Anthropic, and open-source models (Llama, Hugging Face); prompt engineering and systematic LLM evaluation

Programming: Python (primary), TypeScript, JavaScript, SQL, R

Full-Stack & Deployment: React, Node.js, REST APIs; Git-based development and CI/CD workflows; Docker; production deployment on cloud platforms (AWS / GCP)

Data: relational (PostgreSQL), non-relational & vector (pgvector), and time-series / streaming data structures

ML Frameworks & Tools: PyTorch, Hugging Face, Tableau, Streamlit

Leadership & Communication: led multi-author research projects from prototype to production; mentor and teaching assistant across Samsung DS2, LG AI Youth Camp, and SNU courses; experienced guiding non-technical stakeholders in fast-paced settings

Certifications: SQL Developer (SQLD); Engineer Big Data Analysis (빅데이터분석기사); Big-Data Financial Investment Model Research (FIM), SNU & QUANTIT (2023)